# MTH209: Assignment 1

## Debugging assignment

Consider an $n \times n$ square matrix $A$ whose elements are denoted by $a_{ij}$ for $i, j = 1, \dots, n.$ . For any given matrix, we want to find the vector $t \in \mathbb{R}^n$ such that $t = (t_1, t_2, \dots, t_n)$ where each $t_k$ is

$$t_k = \sum_{j=1}^{k} a_{jj} \,.$$

Our goal is to write a function `diagTelescope` that takes a given matrix $A$ and returns the vector $t$ (you may assume that the matrix provided is a square matrix). Below is such a function, but it has errors:

```
diagTelescope <- functin(B)
{
  n <- length(A)
  t < - numeric(length = n)

  alldiags <- diagonal(A)
  for(k in 1 to n)
  {
    t[k] <- sum(alldiags[1:i, ])
  }

  return(t)
}
```

Your job in this assignment is to fix ALL the mistakes and make the function return a valid numerical value.

**INSTRUCTIONS:** copy and paste **ONLY** the function `diagTelescope` in your `assignment1.R` file in the GitHub repository. NO PARTIAL MARKING for this assignment.